

ASSIGNMENT-3.3

Name : kasuri Rohith

Hall no : 2303a510i2

Batch : 23

Task 1: AI-Generated Logic for Reading Consumer Details

Scenario

An electricity billing system must collect accurate consumer data to calculate electricity consumption and charges.

Task Description

An AI tool (GitHub Copilot / Gemini) is used to generate a Python program that performs the following operations:

- Reads Previous Units (PU)
- Reads Current Units (CU)
- Reads Type of Customer
- Calculates the units consumed
- Implements all logic directly in the main program without using functions

Expected Output

- Correct input reading
- Correct calculation of units consumed
- Screenshot showing AI-generated code
- Sample input and output

Prompt Used

Write a Python program for an electricity billing system that reads previous units and current units and type of customer. It should calculate units consumed. The logic should be implemented directly in the main program, and energy charges should depend on the number of units consumed and customer type.

Code:

```
# Task 1: Reading Consumer Details

previous_units = int(input("Enter Previous Units (PU): "))
current_units = int(input("Enter Current Units (CU): "))
customer_type = input("Enter Type of Customer (Domestic/Commercial/Industrial): ")

units_consumed = current_units - previous_units

print(f"Previous Units : {previous_units}")
print(f"Current Units : {current_units}")
print(f"Customer Type : {customer_type}")
print(f"Units Consumed : {units_consumed}")
```

Output:

```
PS C:\Users\rohit\OneDrive\Desktop\ai> & C:/Users/rohit/AppData/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/rohit/OneDrive/Desktop/ai/assign.py
Enter Previous Units (PU): 15
Enter Current Units (CU): 25
Enter Type of Customer (Domestic/Commercial/Industrial): commercial
Previous Units : 15
Current Units : 25
Customer Type : commercial
Units Consumed : 10
PS C:\Users\rohit\OneDrive\Desktop\ai>
```

Task 2: Energy Charges Calculation Based on Units Consumed

Scenario

Energy charges depend on the number of electricity units consumed and the type of customer.

Task Description

The AI-generated code from Task 1 is extended to:

- Calculate **Energy Charges (EC)**
- Use conditional statements based on customer type:
 - Domestic
 - Commercial
 - Industrial
- Improve readability and clarity of logic
- Implement logic directly in the main program (no functions)

Prompt Used

In the above code, energy charges depend on the number of units consumed and customer type.

Now the program should calculate energy charges for domestic, commercial, and industrial consumers.

Code:

```
# Task 2: Energy Charges Calculation
# Original version: direct conditional logic
if customer_type.lower() == "domestic":
    if units_consumed <= 100:
        ec = units_consumed * 1.5
    elif units_consumed <= 200:
        ec = 100 * 1.5 + (units_consumed - 100) * 2.5
    elif units_consumed <= 300:
        ec = 100 * 1.5 + 100 * 2.5 + (units_consumed - 200) * 4
    else:
        ec = 100 * 1.5 + 100 * 2.5 + 100 * 4 + (units_consumed - 300) * 6
elif customer_type.lower() == "commercial":
    ec = units_consumed * 8
elif customer_type.lower() == "industrial":
    ec = units_consumed * 10
else:
    ec = 0
    print("Invalid customer type entered!")
print(f"Energy Charges (EC): ₹{ec:.2f}")
```

Output

```
Enter Previous Units (PU): 15
Enter Current Units (CU): 25
Enter Type of Customer (Domestic/Commercial/Industrial): commercial
Previous Units : 15
Current Units : 25
Customer Type : commercial
```

Task 3: Modular Design Using AI Assistance (Using Functions)

Scenario

The electricity billing logic must be reusable for multiple consumers. To achieve reusability and clarity, the billing calculations are implemented using user-defined functions.

Task Description

Using AI assistance, a Python program is developed that:

- Uses **user-defined functions**
- Calculates:
 - Energy Charges (EC)
 - Fixed Charges (FC)

- Returns calculated values from functions
- Includes meaningful comments for better readability

Prompt Used

In the above code, billing logic must be reusable for multiple consumers. It should use defined functions to calculate energy charges and fixed charges and return the calculated values.

Code:

```
# Task 3: Modular Design (Functions)
def calculate_energy_charges(units, ctype):
    """Calculate EC based on customer type and units consumed."""
    if ctype.lower() == "domestic":
        if units <= 100:
            return units * 1.5
        elif units <= 200:
            return 100 * 1.5 + (units - 100) * 2.5
        elif units <= 300:
            return 100 * 1.5 + 100 * 2.5 + (units - 200) * 4
        else:
            return 100 * 1.5 + 100 * 2.5 + 100 * 4 + (units - 300) * 6
    elif ctype.lower() == "commercial":
        return units * 8
    elif ctype.lower() == "industrial":
        return units * 10
    else:
        return 0
def calculate_fixed_charges(ctype):
    """Return fixed charges depending on customer type."""
    if ctype.lower() == "domestic":
        return 50
    elif ctype.lower() == "commercial":
        return 100
    elif ctype.lower() == "industrial":
        return 150
    else:
        return 0
# Test modular functions
ec = calculate_energy_charges(units_consumed, customer_type)
fc = calculate_fixed_charges(customer_type)
print(f"Energy Charges (EC): ₹{ec:.2f}")
print(f"Fixed Charges (FC): ₹{fc:.2f}")
```

Output:

```
Energy Charges (EC): ₹80.00
Energy Charges (EC): ₹80.00
Fixed Charges (FC): ₹100.00
Fixed Charges (FC): ₹100.00
```

Task 4: Calculation of Additional Charges

Scenario

Electricity bills include multiple additional charges.

Task Description

Extend the program to calculate:

- FC – Fixed Charges
- CC – Customer Charges
- ED – Electricity Duty (percentage of EC)

Use AI prompts like:

- “Add electricity duty calculation”
- “Improve billing accuracy”

Expected Output

- Individual charge values printed
- Correct duty calculation
- Well-structured output
- Verified intermediate results

Prompt:

In above code it should include multiple additional charges like fixed charge, customer charge, and electricity duty.

Code:

```
# Task 4: Additional Charges
# Customer Charges (CC) – flat ₹30
cc = 30
# Electricity Duty (ED) – 5% of EC
ed = 0.05 * ec
print(f"Fixed Charges (FC): ₹{fc:.2f}")
print(f"Customer Charges (CC): ₹{cc:.2f}")
print(f"Electricity Duty (ED): ₹{ed:.2f}")
```

Output:

```
Fixed Charges (FC): ₹100.00
Customer Charges (CC): ₹30.00
Customer Charges (CC): ₹30.00
Electricity Duty (ED): ₹4.00
Electricity Duty (ED): ₹4.00
```

Task 5: Final Bill Generation and Output Analysis

Scenario

The final electricity bill must present all values clearly.

Task Description

Develop the final Python application to:

- Calculate total bill:
- $\text{Total Bill} = \text{EC} + \text{FC} + \text{CC} + \text{ED}$
- Display:
 - o Energy Charges (EC)
 - o Fixed Charges (FC)
 - o Customer Charges (CC)
 - o Electricity Duty (ED)
 - o Total Bill Amount
- Analyze the program based on:
 - o Accuracy
 - o Readability
 - o Real-world applicability

Expected Output

- Complete electricity bill output
- Neatly formatted display
- Sample input/output
- Short analysis paragraph

Prompt:

In above code it should print energy charges, fixed charges, customer charge, electricity duty, total bill amount.

Code:


```

# Task 5: Final Bill Generation
total_bill = ec + fc + cc + ed
print(" Electricity Bill")
print(f"Customer Type    : {customer_type}")
print(f"Units Consumed    : {units_consumed}")
print(f"Energy Charges (EC): ₹{ec:.2f}")
print(f"Fixed Charges  (FC): ₹{fc:.2f}")
print(f"Customer Charges(CC): ₹{cc:.2f}")
print(f"Electricity Duty(ED): ₹{ed:.2f}")
print(f"Total Bill Amount   : ₹{total_bill:.2f}")

```

Output:

```

Enter Previous Units (PU): 15
Enter Current Units (CU): 25
Enter Type of Customer (Domestic/Commercial/Industrial): commercial
Previous Units : 15
Current Units  : 25
Customer Type  : commercial
Units Consumed : 10
Units Consumed : 10
Energy Charges (EC): ₹80.00
Energy Charges (EC): ₹80.00
Fixed Charges (FC): ₹100.00
Fixed Charges (FC): ₹100.00
Fixed Charges (FC): ₹100.00
Customer Charges (CC): ₹30.00
Fixed Charges (FC): ₹100.00
Customer Charges (CC): ₹30.00
Customer Charges (CC): ₹30.00
Electricity Duty (ED): ₹4.00
Electricity Duty (ED): ₹4.00
Electricity Bill
Customer Type    : commercial
Units Consumed   : 10
Customer Type    : commercial
Units Consumed   : 10
Energy Charges (EC): ₹80.00
Fixed Charges  (FC): ₹100.00
Customer Charges(CC): ₹30.00
Electricity Duty(ED): ₹4.00
Total Bill Amount : ₹214.00

```